

CHANDIGARH UNIVERSITY APEX INSTITUTE OF TECHNOLOGY

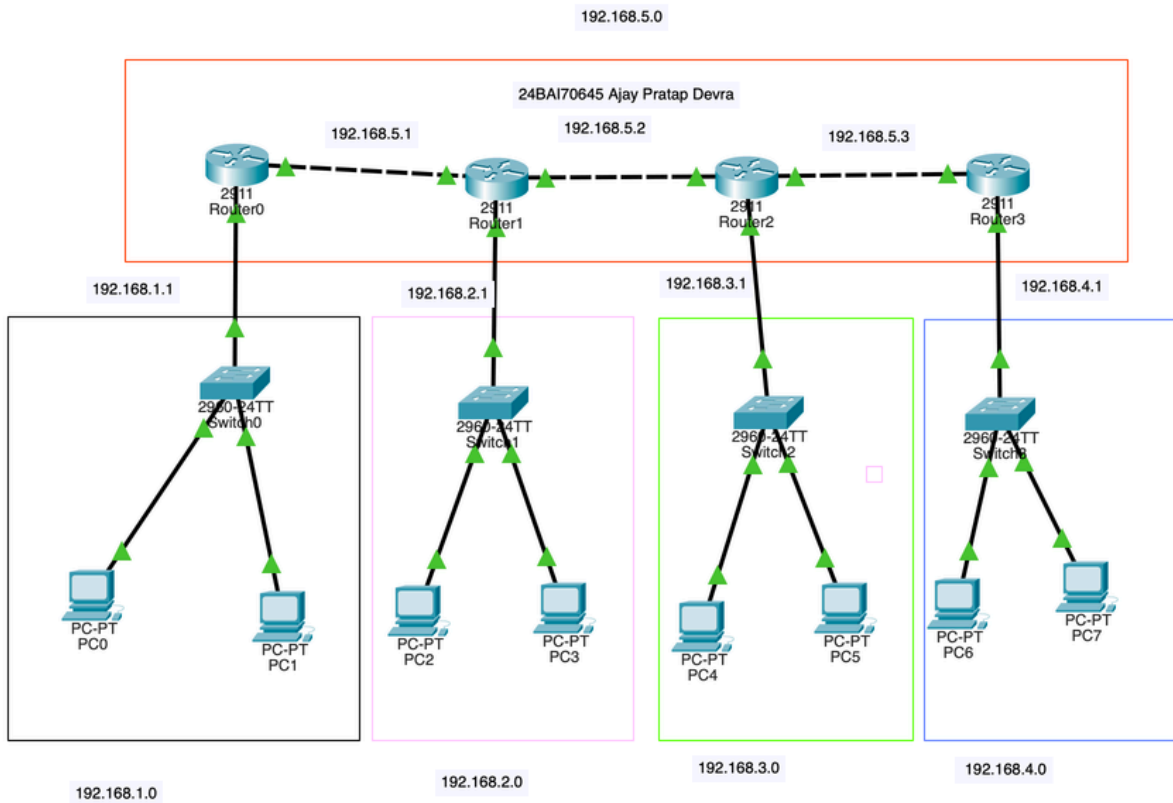


ASSIGNMENT 1

Subject: Computer Networks
Subject Code: 24CST-339

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Q1. Develop Static Routing with 4 PC's, 4 Switches & 8 PC's. Make sure to highlight the Networks with their network ID's. IP Addresses should be Written in text format as well. (6 M)



Network Table :

Network	Purpose	Router/interface
192.168.1.0/24	LAN 1	R0 G0/0 = .1
192.168.12.0/30	R0 ↔ R1	R0 .1, R1 .2
192.168.2.0/24	LAN 2	R1 G0/0 = .1
192.168.23.0/30	R1 ↔ R2	R1 .1, R2 .2
192.168.3.0/24	LAN 3	R2 G0/0 = .1
192.168.34.0/30	R2 ↔ R3	R2 .1, R3 .2
192.168.4.0/24	LAN 4	R3 G0/0 = .1

Routing Tabel :

Router	Destination	Subnet Mask	Next Hop
Router0	192.168.2.0	255.255.255.0	192.168.12.2
	192.168.3.0	255.255.255.0	192.168.12.2
	192.168.4.0	255.255.255.0	192.168.12.2
Router1	192.168.1.0	255.255.255.0	192.168.12.1
	192.168.3.0	255.255.255.0	192.168.23.2
	192.168.4.0	255.255.255.0	192.168.23.2
Router2	192.168.1.0	255.255.255.0	192.168.23.1
	192.168.2.0	255.255.255.0	192.168.23.1
	192.168.4.0	255.255.255.0	192.168.34.2
Router3	192.168.1.0	255.255.255.0	192.168.34.1
	192.168.2.0	255.255.255.0	192.168.34.1
	192.168.3.0	255.255.255.0	192.168.34.1

Ques- What is OSI Model? what are different layers of OSI Model? Write the functionality, Data format and protocols used at each layer.

Answer- The OSI (open system interconnection) Model is a reference model developed by the International Organization for Standardization (ISO) to standardize network communication between different computer systems. It divides the communication process into seven distinct layers, with each layer responsible for specific networking functions.

layer	functionality	Data format	Common protocols
→ Application	Provides network services to end-user applications such as web browsers, email & file data transfer applications.	Data	HTTP, HTTPS, FTP, SMTP, DNS
→ presentation	handles data translation, encryption, decryption and compression to ensure that data is represented in compatible format	Data	SSL/TLS, JPEG, ASCII, MPEG
→ Session	Establishes, manages that and terminates communication sessions between applications	Data	NetBIOS, RPC

	Layer	functionality	Data format	Protocol
→	Transport	Provides end-to-end communication, segmentation, flow control, error control, and reliable data delivery	Segment / Datagram	TCP, UDP
→	Network	performs logical addressing and determines the appropriate path for transmitting packets between different network	Packet	IP, ICMP, OSPF, RIP
→	Data link	Provides node-to-node delivery, framing, MAC addressing and error detection	frame	Ethernet, PPP, HDLC
→	Physical	Transmits raw binary data through physical transmission media such as cables, fiber optics and radio signals	Bits	Ethernet, PHY, USB, DSL, Wi-fi PHY.

Data Encapsulation:

→ During transmission, data pass through the OSI layers from the Application layer to the physical layer. Each layer may add its own control information to the data. The resulting data units are:

Data → segment → Packet → frame → Bits.

At the receiving end, the process is reversed and is

known as decapsulation.

Conclusion: The OSI model provides ~~information~~ a standardized framework for understanding network communication. By dividing communication into Seven layers, it simplifies network design, implementation, troubleshooting and interoperability between different networking systems.

Ques - What are the different Error-detection Methods, Explain each with the help of proper examples.

Answer - During data transmission, errors may occur due to noise, interference signal attenuation or other transmission problems. Errors detection techniques are used to determine whether the data received by the destination has been altered during transmission.

→ The major error detection techniques are:

1. Parity check
2. checksum
3. Cyclic Redundancy Check (CRC)

1. Parity check:-

→ It is one of the simplest error detection techniques. A parity bit is added to the transmitted data so that the total number of binary [1] becomes either even or odd.

Even parity example:

Consider data

→ 1011001

The number of 1s is 4, which is already even. Therefore the parity bit is 0.

Original data : 1011 001

Parity Bit : 0

Transmitted : 10110010

At the receiver, the number of 1s is checked.

If the expected parity condition is not satisfied, an error is detected.

Limitation: A parity check may fail to detect errors when an even number of bits are changed.

2. Checksum:

→ Divides the data into fixed-size blocks and performs binary addition on these blocks. The one's complement of the resulting sum is used as the checksum.

Example:

Consider two 4-bit data block:

1010

1100

1011

The carry is added back to the lower four bits.

0110

+ 1

0111

Taking the One's complement.

0111 → 1000

Therefore the checksum = 1000,

The checksum is transmitted along with the data. At the receiver, the received block and checksum are processed again. Any inconsistency indicates that an error has occurred during transmission.

3. Cyclic Redundancy Check (CRC)

It is widely used and highly reliable error detection technique in computer networks. It uses modulo-2 Binary division Based on a predefined generator polynomial.

Working of CRC :

1. The sender takes the original data.
2. Zeros are appended to the data according to the degree of the generator polynomial.
3. The resulting binary sequence is divided by the generator using modulo-2 division.
4. The remainder obtained from the division is called CRC.
5. This remainder is appended to the original data and transmitted.
6. The receiver divides the received data by the same generator.
7. If the remainder is zero, the data is considered to be free of error, otherwise an error is detected.

Example:

Consider:

Data = 1101

Generator = 1011

The Generator has 4 Bits, so 3 zeroes are appended to the data.

1101000

The sender performs modulo-2 division using XOR operation. The resulting 3-bit transfer remainder is the CRC, which is appended to the original data before transmission.

→ At the receiver, the complete received sequence is divided by the same generator. A non-zero remainder indicates the presence of an error.

Conclusion:

Error detection techniques are essential for maintaining the integrity of transmitted data. Parity checking provides a simple mechanism for detecting basic errors, checksum provides improved detection for blocks of data, and CRC provides a highly reliable error detection and is widely implemented in modern networking systems.